Université d'Alger 1 Benyoucef Benkhedda

Département Informatique



Module: Introduction to Image Processing

Project: A Comparative Study of RAG Techniques for Image Captioning

Phase 1 Report: Data Exploration and Preprocessing

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1 Introduction

This report presents the first phase of our project on Retrieval-Augmented Generation (RAG) for image captioning. The main goal of the project is to see how different RAG techniques can improve the captions generated by Vision-Language Models, making them more accurate and reducing hallucinations on the MS-COCO dataset.

In this document, we focus on Phase 1. Before we can start building the RAG system, we need to make sure our data is clean, well-organized, and ready to use. This phase covers the data exploration, cleaning, splitting, and the creation of the knowledge base that we will use in the next phases.

We worked with a subset of 10,000 images from COCO train2017, along with their reference captions. We also created a validation subset and a completely separate test subset from COCO val2017 to evaluate our future models.

2 Dataset Overview

2.1 Source and Size

- **Knowledge base (training):** 10,000 images from COCO train2017 (containing 50,036 captions).
- **Validation subset:** 100 images from COCO train2017 (subset of the knowledge base).
- **Test subset:** 100 disjoint images from COCO val2017 (stratified, frozen).

2.2 Annotation Format

COCO provides JSON annotations containing image IDs, captions, object bounding boxes, category labels, and segmentation masks. We used these annotations for our statistical exploration, object counting, and data cleaning steps.

Text Cleaning: For retrieval indexing only, captions were lowercased and whitespace was normalized. The original raw captions were kept unchanged for evaluation to avoid data leakage.

3 Statistical Exploration

To better understand our data, we performed a statistical exploration to look at the distribution of visual content and caption properties.

3.1 Object Class Distribution

Figure 1 shows the 20 most frequent object categories (out of the 80 existing classes) in our knowledge base. As requested in the project guidelines, we analyzed the full distribution of the 50 most frequent categories, but only the top 20 are displayed here for readability. As we can see, the distribution is highly unbalanced: "person" is by

far the most frequent category, followed by vehicles (car, motorcycle, truck) and indoor objects (chair, bottle, cup, dining table). This reflects the typical composition of the COCO dataset.

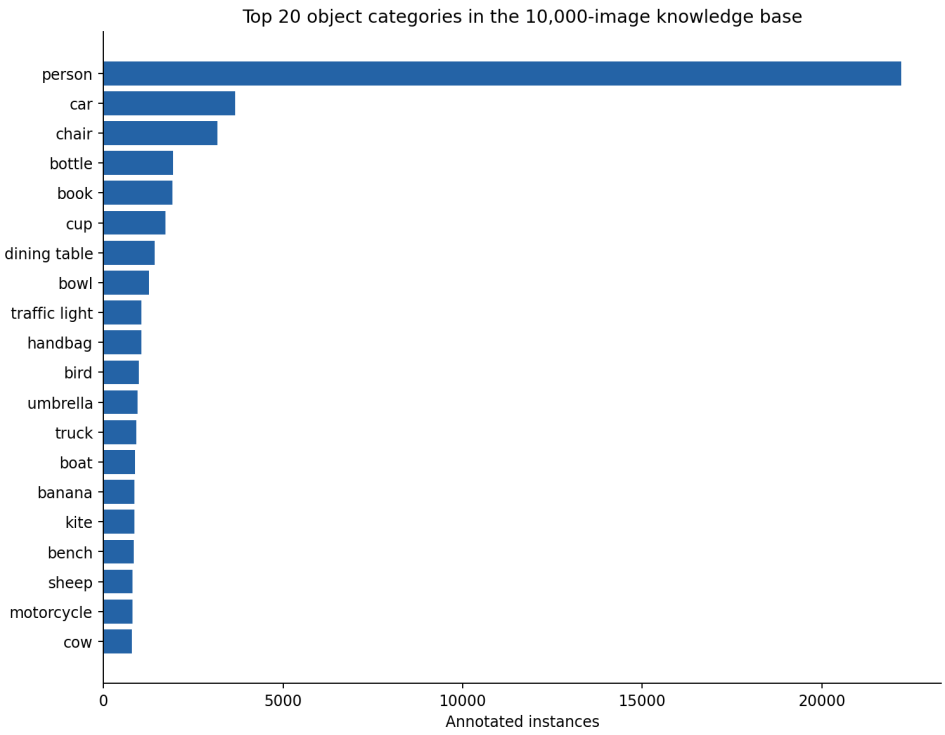


Figure 1: Top 20 object categories in the 10,000-image knowledge base.

3.2 Scene Complexity – Number of objects per image

The number of objects per image was computed using the COCO `instances_train2017.json` annotations. For the final subset of **10,000** images, the average number of objects per image is **7.34** (median = **4.0**, max = **70**).

The distribution is right-skewed, meaning that the majority of images contain a small number of objects (typically between 1 and 5), while a minority of images represent highly complex multi-object scenes. Note that 9,907 images contain at least one annotated object, while a small minority of 93 images have no bounding box annotations (e.g., landscape scenes).

Regarding spatial relationships, the COCO captions frequently describe interactions such as "riding", "holding", or "standing on", which shows that the dataset captures complex object interactions and not just isolated objects.

This heterogeneity in scene complexity is a key finding. It justifies the need for adaptive RAG techniques in Phase 2: visually simple images may require different retrieval strategies than complex, crowded scenes.

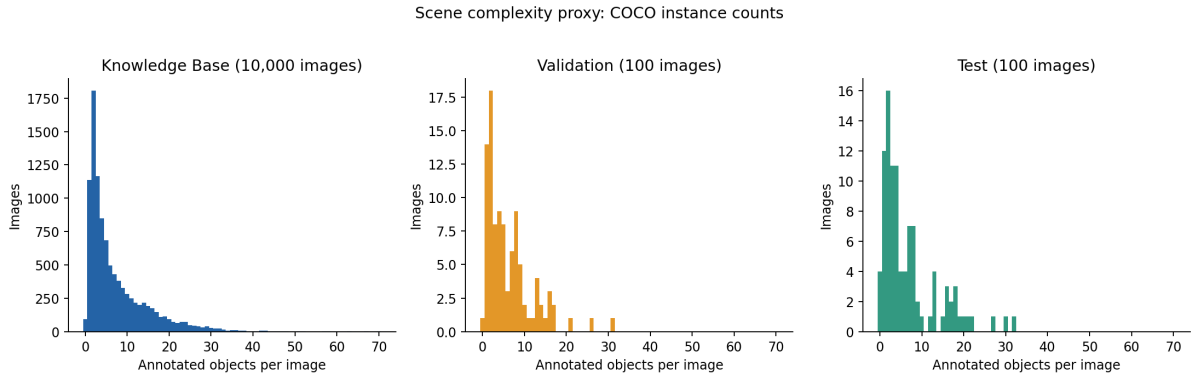


Figure 2: Distribution of the number of objects per image.

3.3 Caption Length Distribution

Figure 3 shows the distribution of the number of words per caption in the dataset. The captions are relatively short and consistent, which is typical for the MS-COCO dataset where each image is described by five independent annotators.

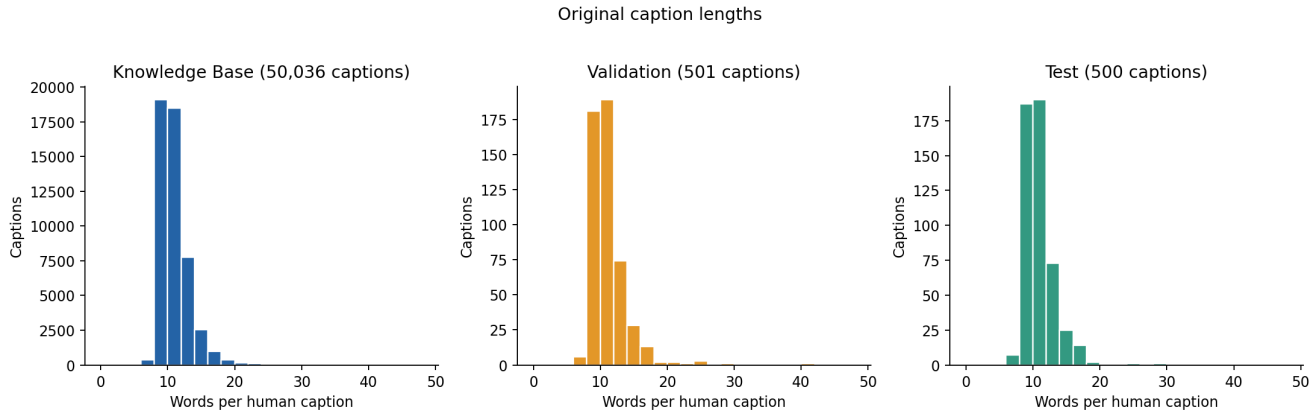


Figure 3: Distribution of caption lengths (number of words).

3.4 Top Words and Vocabulary Analysis

To further understand the linguistic structure of the dataset, we extracted the 30 most frequent content words in the knowledge-base captions (Figure 4). Stop words such as "a", "the" and "on" were removed first, because they would otherwise dominate the ranking without saying anything about the image content. The remaining list is led by "man", "sitting" and "standing", and mixes object nouns (table, street, plate, dog), action verbs (holding, riding, playing) and colour adjectives (white, black, red), which reflects the descriptive nature of the captions. Furthermore, the Zipf plot (Figure 5) confirms that the word distribution follows a power law, which is a standard characteristic of natural language.

After lowercasing and punctuation removal, the knowledge-base captions contain **9,615** unique words, and about 39% of them appear only once.

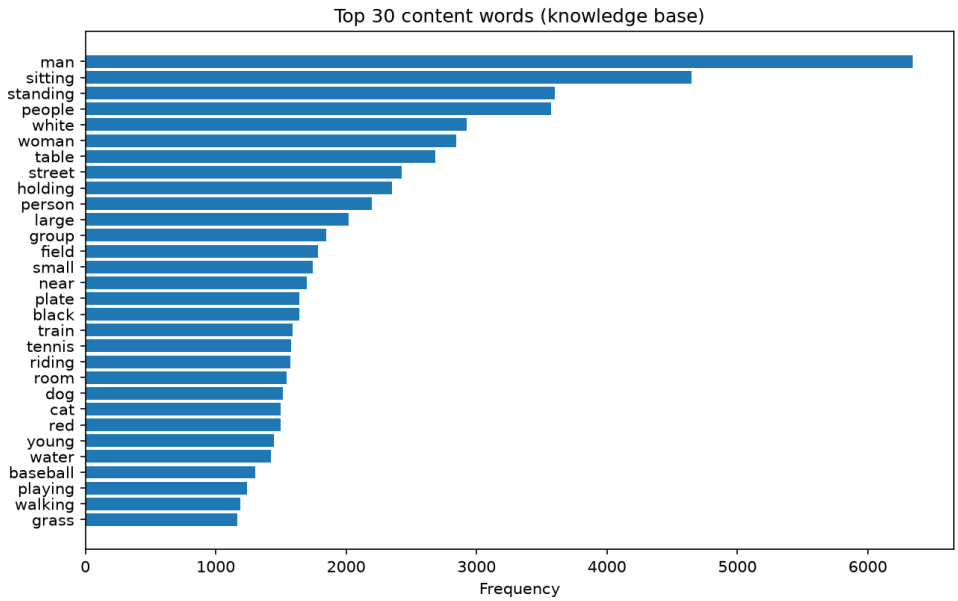


Figure 4: Top 30 content words (stop words removed) in the knowledge-base captions.

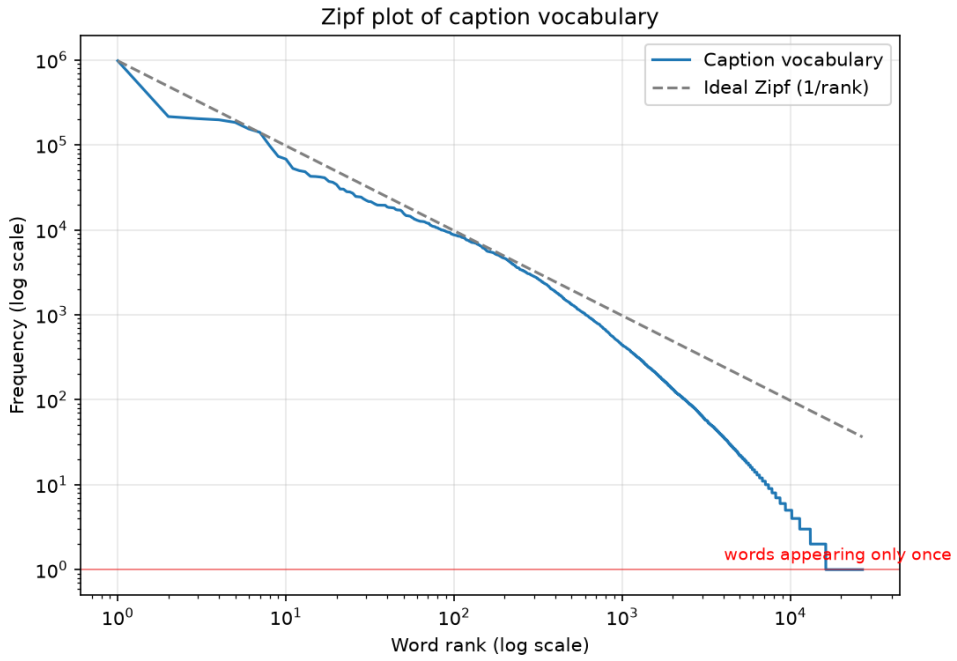


Figure 5: Zipf plot of the caption vocabulary.

3.5 Part-of-Speech (POS) Distribution

The grammatical structure of the captions was analyzed using Part-of-Speech tagging. As shown in Figure 6, nouns and verbs dominate the distribution. This confirms that

the captions are primarily focused on describing who or what is in the image (nouns) and what they are doing (verbs).

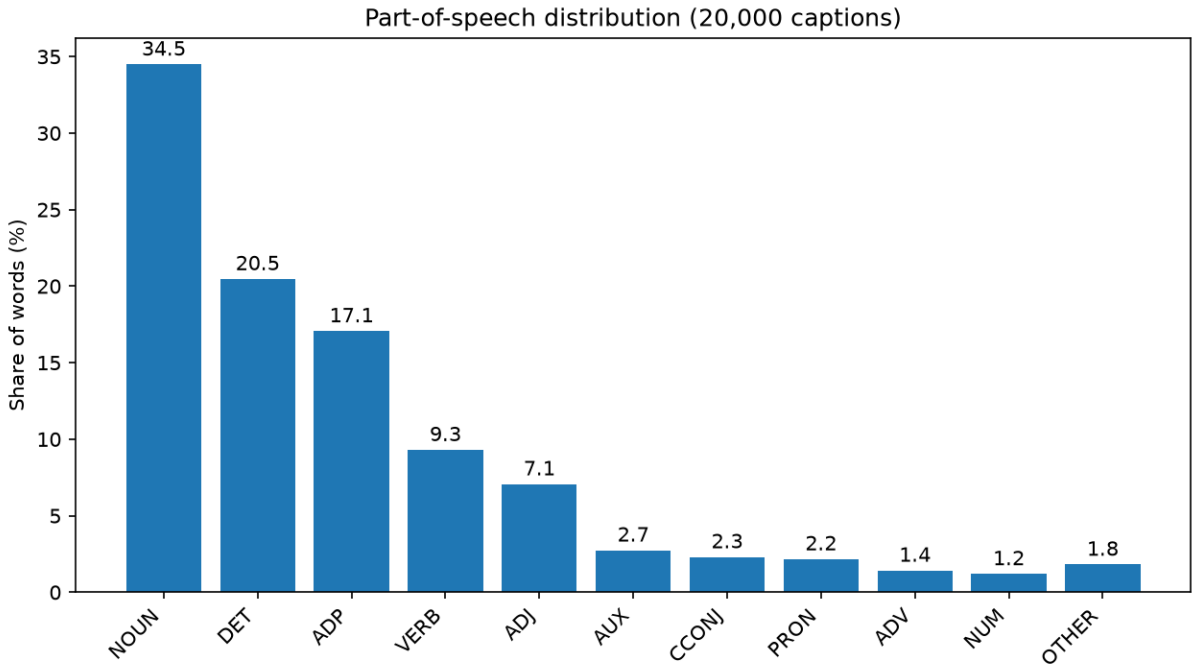


Figure 6: Part-of-Speech distribution in captions.

3.6 Diversity Analysis

Beyond just objects, the dataset contains a rich variety of attributes and contexts. Figure 7 shows the distribution of visual attributes (such as colors), while Figure 8 highlights the diversity of contexts (indoor vs. outdoor, day vs. night).

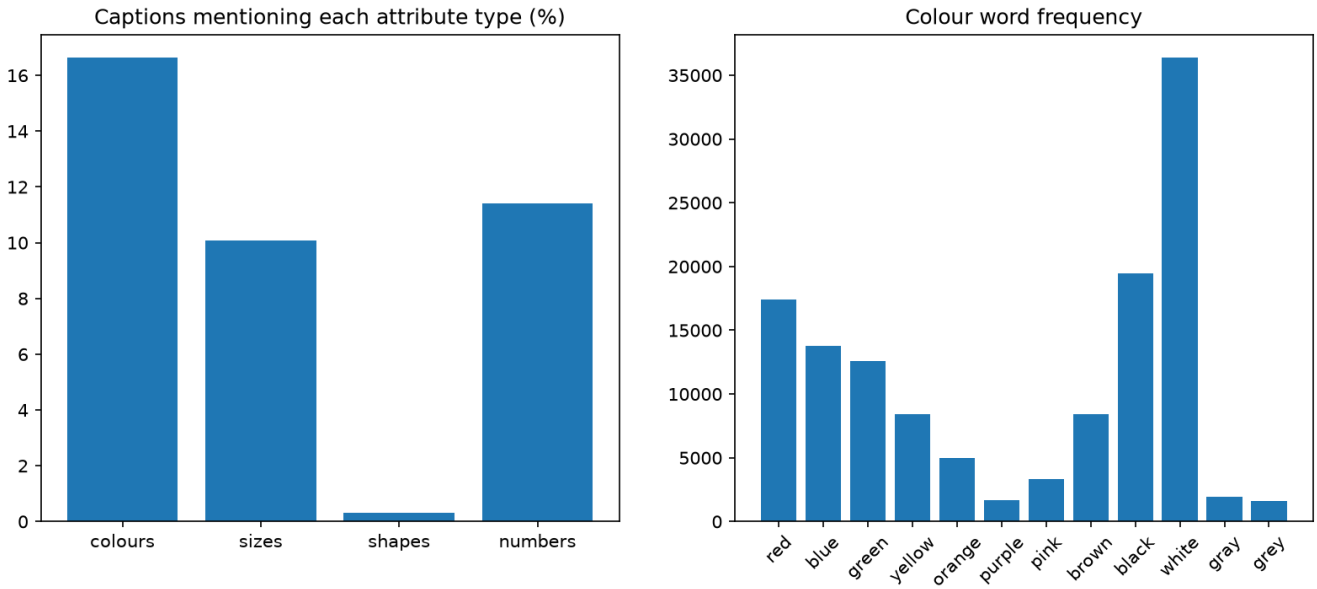


Figure 7: Distribution of attributes in the dataset.

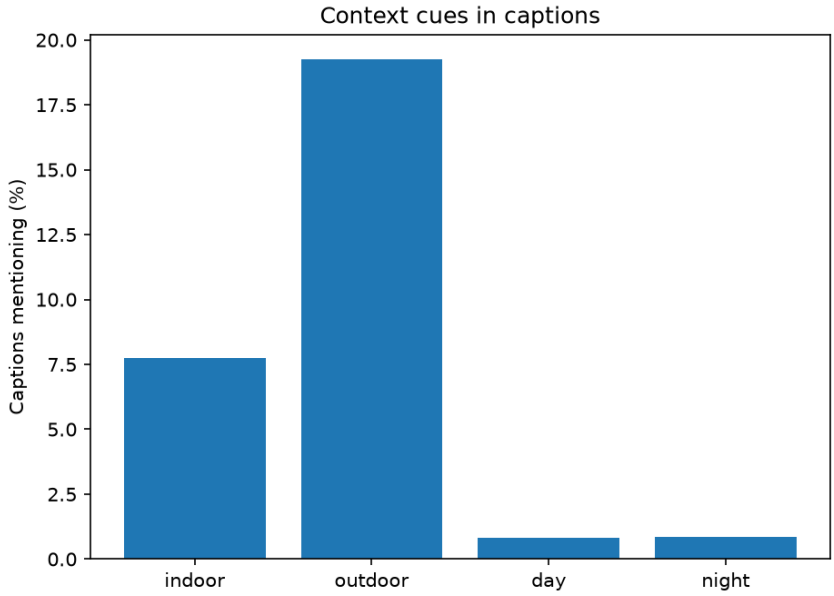


Figure 8: Diversity of contexts in the dataset.

4 Data Cleaning

Before creating our splits, we ran a complete cleaning audit on all 123,287 annotated train2017 and val2017 images. The following operations were applied:

- **Duplicate detection:** Performed using SHA256 hashing of file bytes and decoded RGB pixels. Duplicates were removed, keeping one copy per unique image.
- **Corrupted image detection:** Performed using PIL's `Image.verify()`. Corrupted files were discarded.
- **Missing image detection:** Cross-checked the list of image IDs referenced in the annotations against the actual files present in the image folder.

Results:

- Duplicates removed: **15**
- Corrupted images removed: **0**
- Missing images: **0**
- Final valid unique candidates: **123,272**

From these valid candidates, we sampled **10,000** images to form our knowledge base. The cleaned dataset and updated annotations were stored in the `cleaned_dataset/` folder.

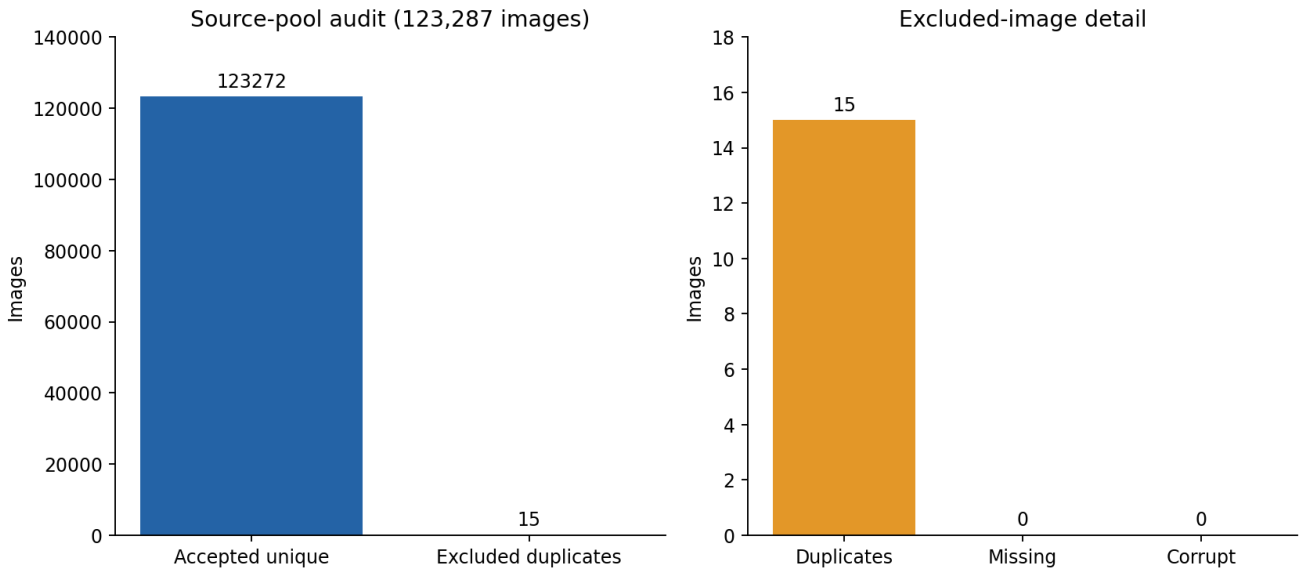


Figure 9: Source-pool audit results and excluded-image details.

5 Data Splits

Three splits were created and frozen for the entire project. The sizes and purposes are detailed in Table 1.

Split	Source	Size	Purpose
Knowledge base (train)	COCO train2017	**10,000**	RAG knowledge base
Validation	COCO train2017	**100**	Hyperparameter tuning
Test	COCO val2017	**100**	Final evaluation

Table 1: Data splits for the project.

To ensure a fair evaluation in Phase 2, we applied strict anti-leakage controls:

- Test image IDs are completely disjoint from the knowledge base.
- Test reference captions are never included in the retrieval index.
- Split IDs are frozen in one `image_ids.json` file per split (`cleaned_dataset/knowledge_base/`, `validation/` and `test/`), generated with a fixed random seed (seed = **42**).

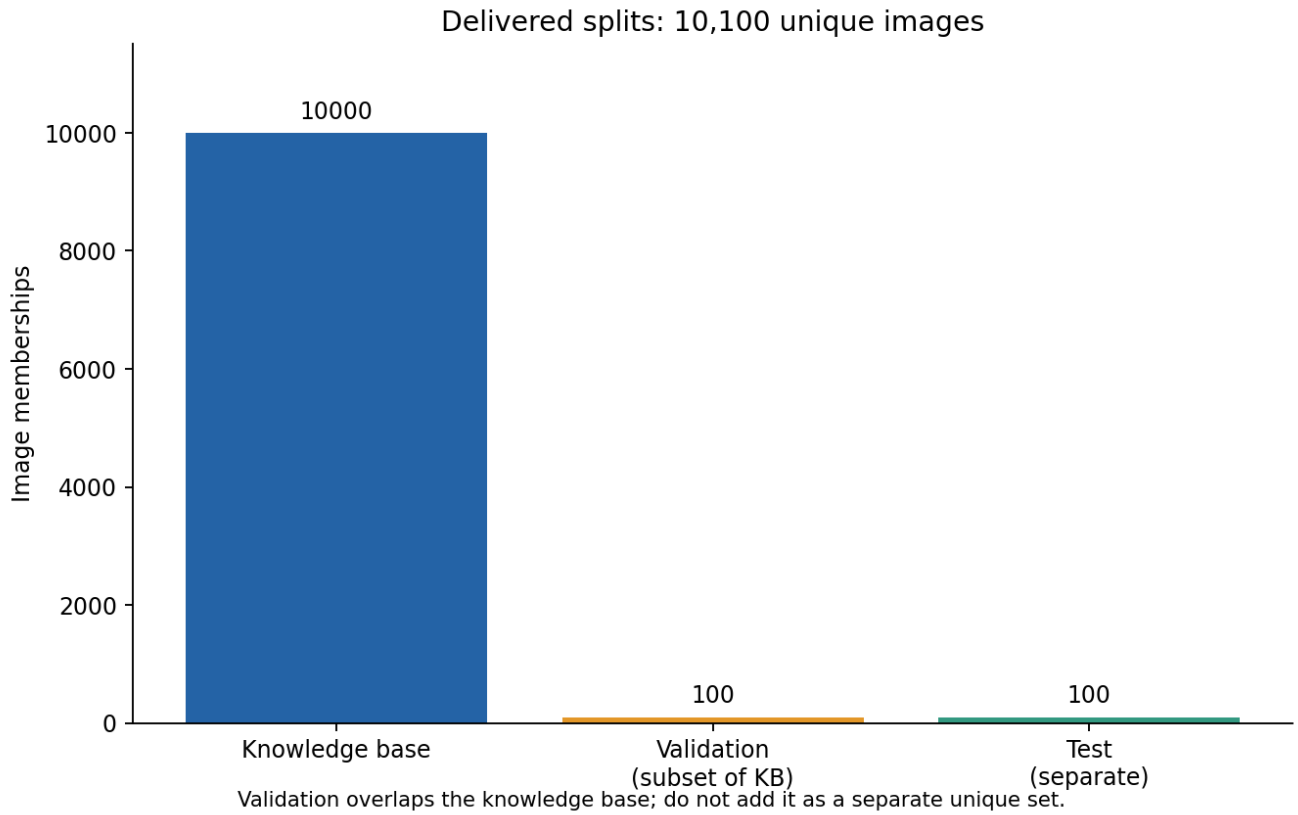


Figure 10: Delivered splits showing the intentional overlap of the validation subset within the knowledge base.

6 Preprocessing and Embeddings

6.1 Image Preprocessing

Images were converted to RGB and resized to meet the CLIP ViT-B/32 encoder requirements (224x224 center crop). EXIF orientation was corrected. Figure 11 shows that original images vary in size, confirming the need for this preprocessing step.

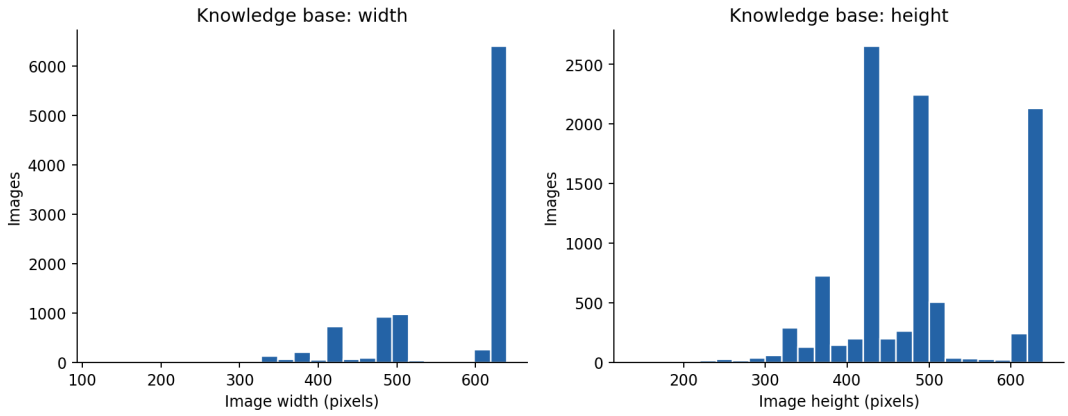


Figure 11: Original image width and height distributions in the knowledge base.

6.2 Embedding Generation

- **Image encoder:** CLIP ViT-B/32 (OpenAI)
- **Embedding dimension:** 512
- **Normalization:** L2
- **Total embeddings: 10,000** images (and 50,036 captions)

Embeddings were saved in `.npy` files and FAISS indexes were built to be used for the retrieval step in Phase 2.

FAISS Documentation: The FAISS index uses exact search (`IndexFlatIP`) with cosine similarity (inner product on L2-normalized vectors). The database size is 10,000 image vectors and 50,036 caption vectors.

Quality Control: We re-encoded the validation images and confirmed that the stored vectors matched with a cosine similarity of ≥ 0.999. No NaNs or duplicate IDs were found.

7 Code and Reproducibility

All the source code, scripts, split files and documentation can be found on our GitHub repository:
https://github.com/Younes769/image-proc

The cleaned image files are too large for GitHub and are hosted on Google Drive:
https://drive.google.com/drive/folders/1xh4MCusgpVPTY2YH83h9KwDsod2Iqn36

8 Conclusion

In this first phase, we successfully produced a cleaned, documented, and frozen dataset of approximately **10,000** images with their reference captions. We split the data into a knowledge base, a validation set, and a test set. We also pre-computed the CLIP embeddings, which are now ready to be used for the retrieval index in Phase 2.

After validating our preprocessing steps on the validation subset, we froze the final dataset and shared it with all group members. We made sure that no modifications will be made to the test split for the rest of the project to keep our evaluations fair.

The main limitations we faced during this phase are:

- The dataset size (10k images) is smaller than the full COCO dataset (123k images), which may limit retrieval diversity.
- Object counts rely on the original COCO annotations, which are known to be incomplete for some scenes.

The dataset is now frozen and fully ready for the RAG-based captioning experiments of Phase 2.

A Appendix: Project Artifacts

To ensure the reproducibility of our work, the code and all lightweight artifacts (split IDs, captions, object annotations, manifests, audit files, statistics and figures) are stored in our GitHub repository. Large binary files are not stored on GitHub because of its file-size limits:

- The cleaned image files (10,200 files, about 1.7 GB) are hosted on Google Drive: https://drive.google.com/drive/folders/1xh4MCusgpVPTY2YH83h9KwDsod2Iqn36
- The embedding matrices (`.npy`) and FAISS indexes (about 380 MB) are regenerated from these images with `build_embeddings.py`. The script `check_embeddings.py` then verifies every file against the SHA256 checksums recorded in `embeddings/metadata.json`.

The repository is organized as follows:

1. Cleaned Dataset (`cleaned_dataset/`)

This folder contains the frozen data splits used for the project. On GitHub, each split includes the image ID list (`image_ids.json`), the filtered captions (`captions.json`), the object annotations (`instances.json`), and a manifest file (`manifest.json`). The Google Drive copy of the same folder also contains the `images/` subfolder of each split.

- `knowledge_base/` : 10,000 images and 50,036 captions used for the RAG database.
- `validation/` : 100 images used for hyperparameter tuning.
- `test/` : 100 disjoint images used for final evaluation.

2. Embeddings (`embeddings/`)

On GitHub, this folder contains the row-to-ID maps (`kb_image_ids.json`, `test_image_ids.json`, `validation_kb_rows.json`), the caption table (`kb_captions.csv`), the build configuration and checksums (`metadata.json`) and the quality-control results (`qc_report.json`). Running `build_embeddings.py` adds the vectors and FAISS indexes used in the retrieval phase:

- `clip_vit_b32/` : CLIP image and caption embeddings (512 dimensions) and their FAISS indexes.
- `minilm_l6_v2/` : Sentence-BERT caption embeddings (384 dimensions) and their FAISS index.

3. Source Code (`.py` scripts and `notebooks/`)

The scripts used to process the data and generate the statistics presented in this report.

- `clean_and_split.py` : Script used for the data audit, duplicate removal, and split generation.

- `verify_and_visualize.py` : Script used to generate the statistical visualizations and the data audit results.
- `notebooks/01_caption_analysis.ipynb` : Notebook used for the caption length, vocabulary, part-of-speech, attribute and context analysis (Sections 3.3 to 3.6).
- `object_counting.ipynb` : Notebook containing the code for Scene Complexity analysis (Section 3.2).
- `build_embeddings.py` : Script used to generate the CLIP and Sentence-BERT embeddings.
- `check_embeddings.py` : Script used for the embedding quality control and the anti-leakage audit.
- `retrieval.py` : Small retrieval interface over the FAISS indexes, used in the next phases.

4. Audit and Reproducibility

- `summary.json` : General summary of the cleaning audit and split sizes.
- `cleaning_audit.csv` : Detailed log of the source image audit (including duplicate reasons and SHA256 hashes).
- `requirements.txt` : List of required Python packages to reproduce the environment.